

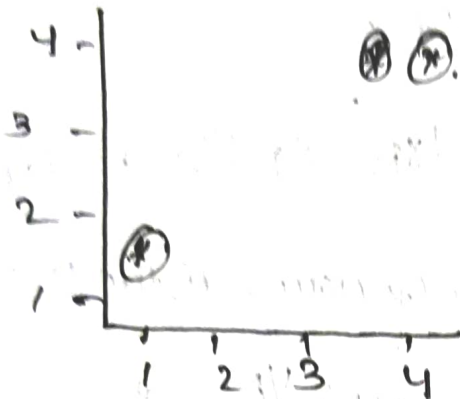
47-5

$$x_1 = (1, 2)$$

$$x_2 = (3, 5)$$

$$x_3 = (3, 0)$$

$$x_4 = (4, 5)$$



manhattan (4)

	x_1	x_2	x_3	x_4
x_1	0			
x_2	5	0		
x_3	3	6	0	
x_4	6	1	7	0

$$(x_1, x_2)$$

$$x_2 = (3, 5)$$

$$\begin{aligned}
 &= |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| \\
 &= 5 - 3 + 5 - 2 \\
 &= 2 + 3 \\
 &= 5.
 \end{aligned}$$

$$\{x_3 = (3, 0), (1, 2)\}$$

$$= |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}|$$

$$= |0 - 3 + 3 - 0$$

$$= 3.$$

$$x_4 = (4, 5)$$

$$= |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}|$$

$$= 5 - 4 + 4 - 1$$

$$= 1 + 3$$

$$= 4.$$

* Euclidean

$$d(i, j) = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \dots + (x_{ip} - x_{jp})^2}$$

	x_1	x_2	x_3	x_4
x_1	0			
x_2	5.61	0		
x_3	2.04	5.1	0	
x_4	4.04	1	5.39	0

* Manhattan

$$d(x_i, y) = \max(|x_1 - y_1|, |x_2 - y_2|)$$

	x_1	x_2	x_3	x_4
x_1	0			
x_2	3	0		
x_3	2	5	0	
x_4	9	1	5	6

$$d(x_1, x_2) = (0, 3) = 3$$

$$d(x_1, x_3) = (2, 2) = 2$$

$$d(x_1, x_4) = (3, 9) = 9$$

$$d(x_2, x_3) = (0, 5) = 5$$

$$d(x_2, x_4) = (1, 1) = 1$$

$$d(x_3, x_4) = \max(1, 5) = 5$$

2) cosine similarity.

find

document	ham coach	hockey	baseball	soccer	p	s	w
document 1	5 0	3	0	2	0	0	2
document 2	3 0	2	0	1	1	0	1
u 3	0 7	0	2	1	0	0	3
u 4	0 1	0	0	1	2	2	0

finding between similarity - blue document 1 and document

$$\cos(d_1, d_2) = (d_1 \cdot d_2) / \|d_1\| \|d_2\|$$

where • vector dot product and length of vector

$$d_1 = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)$$

$$d_2 = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$$

$$d_1 \cdot d_2 = 5 \times 3 + 0 \times 0 + 3 \times 2 + 0 \times 0 + 2 \times 1 + 0 \times 1 + 0 \times 0 + 2 \times 1 + 0 \times 0 + 0 \times 1$$

$$= 5 \times 3 + 3 \times 2 + 2 \times 1 + 2 \times 1 = 25$$

$$\|d_1\| = \sqrt{5 \times 5 + 0 \times 0 + 3 \times 3 + 0 \times 0 + 2 \times 2 + 0 \times 0 + 0 \times 0 + 2 \times 2 + 0 \times 0 + 0 \times 0}$$

$$= \sqrt{25 + 9 + 4 + 4} = \sqrt{42}$$

$$\|dz\| = 3 \times 3 + 0 \times 0 + 2 \times 2 + 0 \times 0 + 1 \times 1 + 0 \times 1 + 0 \times 0 + 1 \times 1 + 0 \times 0 + 1 \times 1$$

$$= 9 + 4 + 1 + 1 + 1 = 17 = 4.12$$